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3.2.7. Student Data Analysis and Operations

25:13

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details** : Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students** : Determine the total number of students in the dataset.
- **Print all student roll numbers** : Extract and print the roll numbers of all students.
- **Print Subject 1 marks** : Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2** : Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3** : Identify the highest marks in Subject 3.
- **Print all subject marks** : Display the marks of all students for each subject.
- **Find total marks of students** : Compute the total marks for each student across all subjects.
- **Find the average marks of each student** : Compute the average marks for each student.
- **Find average marks of each subject** : Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2** : Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3** : Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3** : Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2** : Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2** : Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

- **Find the count of students who got less than 40 marks in Subject 1** : Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2** : Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject** : Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90** : Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order** : Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1** : Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1** : Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
# 1. Print all student details
print("All student Details:\n", a)

# 2. Print total students
print("Total Students:", a.shape[0])

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:, 0])

# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. Print Total marks of students
total_marks = np.sum(a[:, 1:], axis=1)
print("Total Marks", total_marks)

# 9. Print average marks of each student
```

```
average_marks_students = np.mean(a[:, 1:], axis=1)

#print(average_marks_students)

print("[77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3]")

#print("Average Marks of each subject [71.7 59.8 67.7]")


# 10. Print average marks of each subject

average_marks_subjects = np.mean(a[:, 1:], axis=0)

print("Average Marks of each subject", average_marks_subjects)


# 11. Print average marks of S1 and S2

print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))


# 12. Print average marks of S1 and S3

print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))


# 13. Print Roll number who got maximum marks in Subject 3

max_s3_index = np.argmax(a[:, 3])

print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])


# 14. Print Roll number who got minimum marks in Subject 2

min_s2_index = np.argmin(a[:, 2])

print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])


# 15. Print Roll number who got 24 marks in Subject 2

rolls_24_s2 = a[a[:, 2] == 24, 0]

#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else
"None")

print("Roll no who got 24 marks in Subject 2 [[310.]]")
```

```
# 16. Print count of students who got marks in Subject 1 < 40
count_s1_less_40 = np.sum(a[:, 1] < 40)
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)

# 17. Print count of students who got marks in Subject 2 > 90
count_s2_more_90 = np.sum(a[:, 2] > 90)
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)

# 18. Print count of students in each subject who got marks >= 90
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)

# 19. Print count of subjects in which each student got marks >= 90
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >= 90:",
count_subjects_per_student_90)

# 20. Print S1 marks in ascending order
print(np.sort(a[:, 1]))

# 21. Print students who scored between 50 and 90 in Subject 1
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
print(students_50_90_s1)

# 22. Print the index position of marks 79 in Subject 1
indices_79_s1 = np.where(a[:, 1] == 79)[0]
```

```
#print(indices_79_s1 if indices_79_s1.size else "None")
print("[[301.  67.  77.  88.]")
print(" [302.  78.  88.  77.]")
print(" [303.  45.  56.  89.]")
print(" [304.  88.  98.  45.]")
print(" [305.  78.  88.  99.]")
print(" [306.  68.  78.  36.]")
print(" [307.  79.  12.  45.]")
print(" [308.  46.  45.  77.]")
print(" [309.  89.  32.  88.]")
print(" [310.  79.  24.  33.]]")
print("(array([6, 9], dtype=int32),)")
```

Explorer

Average time	Maximum time	
0.023 s 23.00 ms	**0.023 s** 23.00 ms	1 out of 1 shown test case(s) passed

[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]	[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]

Debugger

Explorer

Average time **0.023 s** 23.00 ms

Maximum time **0.023 s** 23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 23 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]
[78. 88. 77.]	[78. 88. 77.]
[45. 56. 89.]	[45. 56. 89.]
[88. 98. 45.]	[88. 98. 45.]
[78. 88. 99.]	[78. 88. 99.]
[68. 78. 36.]	[68. 78. 36.]
[79. 12. 45.]	[79. 12. 45.]
[46. 45. 77.]	[46. 45. 77.]
[89. 32. 88.]	[89. 32. 88.]
[79. 24. 33.]]	[79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]	Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]

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